

# A Connectome Test of the Fly Hashing Algorithm

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**Abstract**—The *Drosophila* olfactory circuit has been modelled as a locality-sensitive hash: glomerular input is expanded through a sparse binary projection onto Kenyon cells and sparsified by winner-take-all. The projection in that model was drawn at random. We replace it with the measured glomerulus-to-Kenyon-cell connectivity of the MaleCNS connectome, released in 2026, and evaluate nearest-neighbour retrieval on synthetic linear mixtures of partially observed DoOR odorant profiles. The measured wiring is structured: glomerulus co-occurrence exceeds that of all 200 uniform degree-preserving rewirings. Its retrieval is slightly worse than theirs: at the pre-specified hash size it scores below 199 of the 200 ( $-2.06\%$ , two-sided  $p = 0.020$ ; 90% bootstrap interval  $-3.1\%$  to  $-0.8\%$ ), inside a pre-specified  $\pm 5\%$  equivalence margin. The deficit shrinks to between  $-0.7\%$  and  $+0.6\%$  when unmeasured responses are imputed rather than set to zero, so it cannot be separated from the missing-data model. Against Gaussian sign codes the fly hash leads only when codes are compared at  $k$  bits; it trails at matched storage, and at matched computation it trails for small  $k$  and is near parity for large  $k$ .

**Index Terms**—connectomics, locality-sensitive hashing, similarity search, null models

## I. INTRODUCTION

Locality-sensitive hashing (LSH) maps items to short codes under which similar items collide with high probability [1], [2]. Dasgupta, Stevens and Navlakha [3] observed that the fly olfactory circuit computes a related transform that departs from classical practice: a sparse binary projection that expands rather than reduces dimensionality, followed by winner-take-all. At matched computational cost it outperformed classical LSH on standard benchmarks. The projection was sampled at random, anchored to biology only by the number of inputs per Kenyon cell, following the then-prevailing account that this connectivity is effectively random [4].

That account has since been revised. Zheng *et al.* [5] mapped projection-neuron input to Kenyon cells in a whole-brain electron microscopy volume, found it structured, and reported that simulations of the observed network discriminated odours worse than a random network, except when signal was restricted to overconvergent, largely food-responsive inputs. The MaleCNS connectome [6] now provides a complete central nervous system of a second animal. We ask a narrower question than [5]: on the retrieval benchmark that motivated the hashing interpretation, does substituting measured connectivity for the random projection change retrieval quality? Our contributions are:

- the fly hashing benchmark evaluated under MaleCNS connectivity against a uniform fixed-margin null, which isolates the pairing of glomeruli with Kenyon cells from both degree sequences;

- a comparison with Gaussian sign codes at three explicit budgets (code length, storage, computation);
- an inference protocol stated in advance of this analysis: an exact randomization test with a pre-specified primary hash size, a two-stage bootstrap whose coverage we check by simulation, a pre-specified equivalence margin, and a power analysis.

## II. METHODS

### A. Projection

Let  $M \in \{0,1\}^{d \times m}$  with  $M_{ji} = 1$  iff glomerulus  $j$  connects to Kenyon cell  $i$ . We build  $M$  from MaleCNS v1.0 using every reconstructed connection, with no synapse-count threshold. Projection neurons of class ALPN are assigned to glomeruli by cell type; multiglomerular, unnamed, thermo- and hygro-sensory types are excluded. For the right hemisphere this gives  $d = 51$  glomeruli and  $m = 1,886$  Kenyon cells. The odour data cover 35 of these glomeruli, and all analyses use that restriction:  $d = 35$ ,  $m = 1,838$ , 7,110 non-zero entries.

We report *distinct glomerular inputs* per cell,  $c_i = \sum_j M_{ji}$  (mean 3.87; 5.28 before restriction), and *fan-out*  $f_j = \sum_i M_{ji}$  (range 19–473, maximal for DC1). Because projection neurons are aggregated by glomerulus,  $c_i$  is not an anatomical claw count; the number of distinct projection-neuron partners is larger (mean 5.59), and neither quantity is validated against claw morphology here.

### B. Hash, baselines and budgets

As in [3], an input  $x \in \mathbb{R}_{\geq 0}^d$  is normalised to unit mean,  $\tilde{x} = x/\bar{x}$ , expanded as  $y = M^\top \tilde{x}$ , and the  $k$  largest entries of  $y$  are set to one. Cells with identical inputs tie exactly. Our rule selects exactly  $k$  winners, breaking ties at the threshold by a fixed random priority over cells; the lowest-index rule and a rule admitting every tied cell are sensitivity analyses. Boundary ties affect at most 2.3% of queries at any  $k$ .

The baseline is a Gaussian sign code,  $\text{sign}(\tilde{x}^\top W)$  with  $W \in \mathbb{R}^{d \times b}$  i.i.d. normal. A  $k$ -sparse tag over  $m$  cells is not a  $k$ -bit code, so we compare at three budgets:  $b = k$  bits (code length equal to the number of active cells);  $b = \lceil \log_2 \binom{m}{k} \rceil$ , the length of an optimal code for an arbitrary  $k$ -of- $m$  tag (storage-matched, 523 bits at the primary  $k$ ); and  $b = \text{nnz}(M)/d = 203$ , equal numbers of input-weight operations per item (computation-matched; normalisation, selection and thresholding excluded on both sides).

### C. Null and comparison matrices

Only  $M$  varies between conditions. The null of record is uniform over binary matrices with the same row and column

sums as  $M$ , sampled by curveball trades [7], which Carstens proved converge to the uniform distribution [8]. We use 30 trade attempts per cell. In five chains run to 100 sweeps, the glomerulus co-occurrence statistic  $Q = \sum_{j < l} (MM^\top)_{jl}^2$  stays within three null SDs from 1 sweep on, the Jaccard overlap with the measured matrix levels off at 0.088 by 10 sweeps, and retrieval scores at 30 and 100 sweeps are indistinguishable (supplement). Comparison matrices are not nulls and each changes something the null keeps: input counts kept but glomeruli uniform (fan-out flattened); six uniform inputs per cell (the construction of [3]); input counts kept with fan-out made uniform and pairing random; and a degraded matrix in which all cells draw from four glomeruli.

### D. Benchmark

Responses are the DoOR 2.0 consensus matrix [9] (commit db323a4), mapped from receptor to glomerulus. Keeping glomeruli measured for  $\geq 40$  odorants and odorants measured on  $\geq 12$  of them leaves 172 odorants, of which 28.7% of entries are unmeasured. These are set to zero in the baseline; unmeasured is not the same as non-responsive, so three imputations are sensitivity analyses. The benchmark consists of 4,000 *synthetic linear mixtures of partially observed DoOR profiles* (Dirichlet weights over two to five sources). They are not measured mixture responses and add no biological measurements; they characterise this generator. Ground truth is the  $\kappa = 10$  nearest items by Euclidean distance on raw responses, a target the concentration-invariant hash cannot fully recover; distances on normalised input and angular distance are sensitivity analyses. Items are ranked by Hamming distance with ties broken by a fixed random order per query, identical across matrices, and scored by mean average precision (mAP).

### E. Inference

The primary hash size, fixed before this analysis, is  $k = 92$ , i.e. 5% of Kenyon cells as in [3];  $k \in \{8, 16, 32, 64\}$  are secondary and Holm-adjusted. Two questions are separated. *Is the measured matrix unusual among uniform fixed-margin matrices?* Under that null hypothesis its score is exchangeable with  $B = 200$  null scores, so the rank gives an exact randomization  $p$ -value conditional on the benchmark. *How large is the difference, across odorant sets?* The estimand is  $\theta = \mathbb{E}_D[s_D(M) - \mathbb{E}_0 s_D] / \mathbb{E}_D \mathbb{E}_0 s_D$ , where  $D$  is a benchmark generated from the source odorants and  $\mathbb{E}_0$  is over null matrices. A two-stage bootstrap resamples source odorants, regenerates mixtures and ground truth, recomputes retrieval, and re-estimates the null mean from 20 freshly drawn null matrices (200 replicates), so the interval carries benchmark and finite-null uncertainty. We check its coverage by simulation from a known population fitted to the data. We distinguish variation across random networks (null SD) from uncertainty in the null mean ( $\text{SD}/\sqrt{B}$ ).

Equivalence is assessed at a margin of  $\pm 5\%$  relative mAP, fixed before this analysis but after an earlier version had been seen; results at  $\pm 2.5\%$  and  $\pm 10\%$  are in the supplement. The margin is anchored to the effects at issue in [3], improvements

TABLE I  
MEASURED WIRING AGAINST THE NULL ( $B = 200$ ) AND AGAINST COMPARISON MATRICES (RELATIVE TO THE NULL MEAN, %). \*PRIMARY.

| $k$ | real   | null                | $\Delta\%$ | $p$   | $p_{\text{Holm}}$ | fan flat | 6 unif. | balanced |
|-----|--------|---------------------|------------|-------|-------------------|----------|---------|----------|
| 8   | 0.1973 | 0.2034 $\pm$ 0.0085 | -3.0       | 0.468 | 1.000             | +3.7     | +26.7   | +3.2     |
| 16  | 0.2879 | 0.2933 $\pm$ 0.0062 | -1.8       | 0.438 | 1.000             | +2.7     | +9.7    | +3.2     |
| 32  | 0.3531 | 0.3575 $\pm$ 0.0045 | -1.2       | 0.338 | 1.000             | +3.3     | +3.0    | +3.6     |
| 64  | 0.3963 | 0.4001 $\pm$ 0.0039 | -0.9       | 0.348 | 1.000             | +3.4     | +0.7    | +3.4     |
| 92* | 0.4063 | 0.4148 $\pm$ 0.0035 | -2.1       | 0.020 | —                 | +3.2     | +0.1    | +3.2     |

TABLE II  
MAP OF THE FLY HASH AND OF GAUSSIAN SIGN CODES (BITS IN PARENTHESES).

| $k$ | fly   | Gauss. $k$ bits | storage-matched | computation-matched |
|-----|-------|-----------------|-----------------|---------------------|
| 8   | 0.197 | 0.019           | 0.264 (72)      | 0.382 (203)         |
| 16  | 0.288 | 0.069           | 0.340 (130)     | 0.382 (203)         |
| 32  | 0.353 | 0.151           | 0.394 (229)     | 0.382 (203)         |
| 64  | 0.396 | 0.249           | 0.436 (397)     | 0.382 (203)         |
| 92  | 0.406 | 0.296           | 0.454 (523)     | 0.382 (203)         |

of up to  $3\times$ : a wiring effect below it could not overturn the modelling assumption being tested. Power is estimated under a location-shift model using the empirical null distribution.

## III. RESULTS

### A. The measured wiring is structured

Glomerulus co-occurrence  $Q$  of the measured matrix is +3.7% above the null mean and exceeds all 200 null draws (upper-tail  $p = 0.005$ ). The pairing of glomeruli within Kenyon cells is therefore not uniform given the degrees, in agreement with [5] in a second animal.

### B. Retrieval is slightly below the null

At the primary size the measured matrix scores 0.4063 against a null of  $0.4148 \pm 0.0035$  (SD across networks; SE of the mean 0.00025), a relative difference of -2.06%. It lies below 199 of the 200 null draws: two-sided randomization  $p = 0.020$ . At the secondary sizes the differences are -3.0% to -0.9%, with unadjusted  $p$  from 0.338 to 0.468 and none significant after Holm adjustment (Table I, Fig. 1). The sign is the same at every size, but the sizes share one benchmark and one matrix, so this is not independent evidence.

The two-stage 90% interval for  $\theta$  at the primary size is -3.1% to -0.8%: the deficit is unlikely to be zero, and equally unlikely to exceed a few percent. Equivalence at the pre-specified  $\pm 5\%$  margin holds; at  $\pm 2.5\%$  it does not hold. In simulation the procedure covered the true value in 100% of 100 datasets (Wilson 95%: 96–100%) at nominal 90%, so it is, if anything, conservative; the group bootstrap of an earlier version of this analysis covered it in 87% (79–92%). The randomization test has 80% power against shifts of 2.5% downward and 2.5% upward.

### C. The architecture comparison depends on the budget

At  $b = k$  bits the fly hash is ahead at every size (Table II), by a factor that falls with  $k$  ( $1.37\times$  at the primary size). At matched storage it is behind at every size ( $0.90\times$ ). At

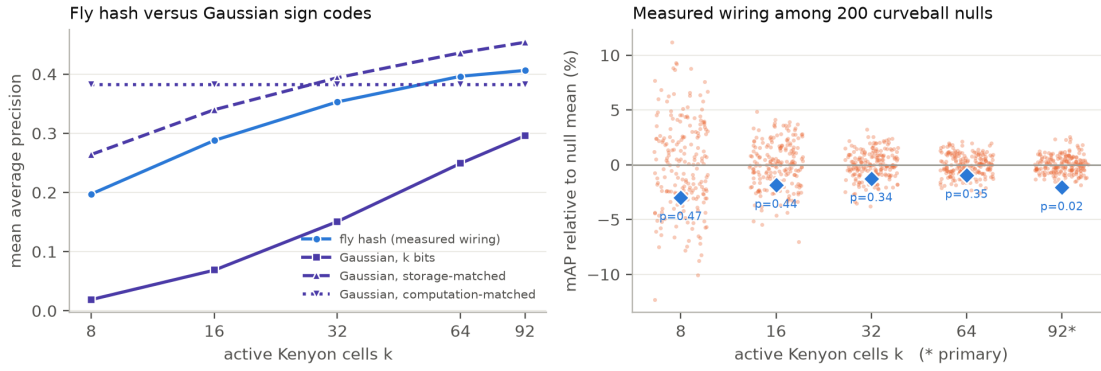


Fig. 1. **Left:** mAP of the fly hash on measured wiring and of Gaussian sign codes at three budgets (mean of 20 seeds). **Right:** each orange point is one of 200 uniform fixed-margin rewirings, blue diamonds the measured wiring, both relative to the null mean; two-sided randomization  $p$  shown.

TABLE III  
PRIMARY SIZE UNDER ONE-FACTOR VARIATIONS ( $B = 40$  NULLS EACH):  
RELATIVE DIFFERENCE TO THE NULL MEAN (%) AND TWO-SIDED  $p$ .

| variation                           | $\Delta\%$   | $p$         |
|-------------------------------------|--------------|-------------|
| Baseline                            | -2.2         | 0.049       |
| Left hemisphere                     | -1.1         | 0.244       |
| $\geq 5$ -synapse threshold         | -2.3         | 0.098       |
| Missing data (3 imputations)        | -0.7 to +0.6 | 0.439–0.780 |
| Coverage thresholds (2)             | -1.7 to -1.4 | 0.049–0.341 |
| Mixture seeds (4)                   | -2.4 to -1.6 | 0.049–0.244 |
| Mixture size 2 / 2–10               | -2.0 to -0.9 | 0.146–0.195 |
| Measured odorants only              | -3.9         | 0.049       |
| Normalised / angular metric         | -2.4 to -1.7 | 0.049       |
| Tie rules (index, include, 2 seeds) | -2.4 to -2.1 | 0.049       |
| Synapse-weighted                    | -1.1         | 0.488       |
| Synthetic inputs (3)                | -3.1 to +0.8 | 0.049–0.244 |

matched computation it is behind for small  $k$  and slightly ahead from  $k = 64$  ( $1.06\times$  at the primary size). The same ordering holds under normalised and angular ground truth. The  $3\times$  advantage reported in [3] was at matched computation on other benchmarks and larger inputs; we do not reproduce its magnitude at the circuit’s own scale.

#### D. Robustness

Across 25 variations (Table III) the relative difference at the primary size ranges from -3.9% to +0.8% and is negative in 23. In 13 the measured matrix falls below all 40 nulls, the smallest attainable  $p$  (0.049); the variations are not independent, and no adjustment is applied. The exception that matters is missing data: imputing unmeasured responses from glomerulus means, odorant means or a rank-5 fit instead of zero gives -0.7% to +0.6%, with  $p \geq 0.439$ . Every variation, with all secondary sizes, is in the supplement.

#### IV. DISCUSSION

The measured pairing is structured, and under the baseline treatment of the data it retrieves slightly worse than uniform rewirings with the same degrees. The direction agrees with [5], whose observed network discriminated odours worse than a random one. The size of the effect, about two percent with an interval reaching three, is small beside the architectural effects

at issue in [3], and the pre-specified equivalence test holds at  $\pm 5\%$ . For this task the random-matrix idealisation therefore loses nothing that the measured pairing provides, and may gain a little.

Two cautions limit that reading. First, the deficit depends on treating unmeasured DoOR entries as zero; with any of three imputations it is no longer detectable. The wiring and the missing-data model interact, and this experiment cannot say which imputation is closer to the truth. Second, the comparison matrices answer different questions from the null because each changes a degree sequence. Flattening fan-out while keeping input counts raises mAP above the null at every size (Table I); that is an intervention on fan-out, and it does not explain the measured matrix’s position relative to its own null, which holds fan-out fixed. That the structure serves other ends, such as innate valence or the prioritised channels of [5], is a hypothesis this experiment does not test.

#### V. LIMITATIONS

One animal, one hemisphere as the primary analysis, one task and one metric. The benchmark is generated: mixtures are linear and synthetic, 28.7% of source entries are unmeasured, and 35 of 51 glomeruli are covered. The model is binary and rate-free, and ignores inhibition beyond winner-take-all. Input-dependence of the kind reported in [5] is untested, as is novelty detection [10].

#### VI. CONCLUSION

In a binary glomerulus-to-Kenyon-cell model evaluated on synthetic mixtures derived from DoOR responses, measured connectivity produced slightly lower retrieval scores than the mean of uniform degree-preserving randomizations, although its pairing is itself non-random. The deficit is small, within a pre-specified equivalence margin, and dependent on how unmeasured responses are treated. The specific measured pairing is not necessary for strong performance on this benchmark; its biological significance requires further evaluation. Code, data checksums, results and a supplement are at <https://github.com/ssenge/FlyHash-Connectome>.

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## REFERENCES

- [1] P. Indyk and R. Motwani, “Approximate nearest neighbors: towards removing the curse of dimensionality,” in *Proc. 30th ACM STOC*, 1998, pp. 604–613.
- [2] M. S. Charikar, “Similarity estimation techniques from rounding algorithms,” in *Proc. 34th ACM STOC*, 2002, pp. 380–388.
- [3] S. Dasgupta, C. F. Stevens, and S. Navlakha, “A neural algorithm for a fundamental computing problem,” *Science*, vol. 358, no. 6364, pp. 793–796, 2017.
- [4] S. J. C. Caron, V. Ruta, L. F. Abbott, and R. Axel, “Random convergence of olfactory inputs in the *Drosophila* mushroom body,” *Nature*, vol. 497, pp. 113–117, 2013.
- [5] Z. Zheng *et al.*, “Structured sampling of olfactory input by the fly mushroom body,” *Current Biology*, vol. 32, no. 15, pp. 3334–3349, 2022.
- [6] S. Berg, I. R. Beckett, M. Costa, P. Schlegel, M. Januszewski, E. C. Marin, *et al.*, “Sexual dimorphism in the complete *Drosophila* male central nervous system connectome,” *Cell*, 2026, doi:10.1016/j.cell.2026.08.015.
- [7] G. Strona, D. Nappo, F. Boccacci, S. Fattorini, and J. San-Miguel-Ayanz, “A fast and unbiased procedure to randomize ecological matrices with fixed row and column totals,” *Nature Communications*, vol. 5, art. 4114, 2014.
- [8] C. J. Carstens, “Proof of uniform sampling of binary matrices with fixed row sums and column sums for the fast Curveball algorithm,” *Physical Review E*, vol. 91, art. 042812, 2015.
- [9] D. Münch and C. G. Galizia, “DoOR 2.0 — comprehensive mapping of *Drosophila melanogaster* odorant responses,” *Scientific Reports*, vol. 6, art. 21841, 2016.
- [10] S. Dasgupta, T. C. Sheehan, C. F. Stevens, and S. Navlakha, “A neural data structure for novelty detection,” *Proc. Natl. Acad. Sci. USA*, vol. 115, no. 51, pp. 13093–13098, 2018.